

What are we discussing?

Hogan asked two things. First: tell him beforehand which mechanisms you are thinking of including, as bullets, not another document. Second: put a link under each point so he can look.

This note is for you. It is the conversation, not the email. Each paper below is about 200 words: the question, the evidence, and the move.

Nothing here changes Table 2. Nothing here rewrites Hogan's HKO paragraph. Confirmatory freeze is still a human decision. All twelve Model 1 q -values are above 0.19 (minimum 0.192). CHD hot nights: 1.022 per five extra official nights. HF cold days: 1.073 per five extra official cold days.

On the table	Move	Paper
Keep overnight recovery after hot nights?	Keep as a hypothesis	Chevance 2024; Ashe 2025 (limit)
Keep cold-related afterload on a failing heart?	Keep as a hypothesis	Ikäheimo 2018; Li 2026
Keep the seven-man heat chamber?	Drop	Ioannou 2024
Keep bedroom temperature and heart-rate variability?	Drop	O'Connor 2025
Infer indoor temperature from outdoor counts?	No	not a paper
Treat five extra official days as a consecutive spell?	No. 2018: 26 official hot nights, zero days in five-night spells	not a paper
Evaluate Hong Kong's heat or cold warnings?	No. Do not count admissions averted	not a paper

Paper 1. Chevance et al. 2024 — keep

The question. Can a month with more official hot nights be different from a month that is merely warm on average? Sleep needs a falling core temperature at night. If nights stay hot, recovery can fail. The discussion is whether a *count of hot nights* is a different monthly question from mean temperature.

What they found. Chevance, Minor, Vielma and colleagues reviewed ambient heat and sleep. Heat, outdoors or indoors, is associated with shorter sleep and poorer sleep quality. Fragmentation is part of that quality story. The paper is a sleep review. It is not a Hong Kong hospital study. It is not people with diabetes or hypertension. It does not measure first CHD hospitalisation.

What our file has. We count official hot nights ($T_{\min} \geq 28^{\circ}\text{C}$ at Headquarters). We do not see sleep. CHD sits at 1.022 per five extra such nights; monthly mean temperature for CHD sits near 1. Overnight recovery is a hypothesis for that contrast. It is not a sleep coefficient.

What we do. Keep Chevance next to the overnight-recovery sentence. Do not import it as a finding.

<https://doi.org/10.1016/j.smr.2024.101915>

Paper 2. Ashe et al. 2025 — keep as a limit

The question. If heat disturbs sleep, can we say sleep is the path to hospitalisation? That would be a mediator claim. Hogan should not be asked to treat it as proven.

What they found. Ashe, Wozniak, Conner and colleagues scoped the literature on extreme heat, sleep, and cardiovascular measures. They found three eligible studies. Blood pressure did not move one way. No study showed that sleep mediates a heat–hospitalisation path. Mixed, thin, not a dipping theorem. Three studies are not a pathway proof. Heat can disturb sleep. That is not the same as showing that sleep carries heat to hospital.

What our file has. The live Discussion already uses this as a refusal: the panel cannot treat sleep as a measured mediator. We have no sleep variable. We have no blood pressure. We have monthly outdoor counts and first-event hospital totals.

What we do. Keep Ashe so the overnight-recovery sentence cannot be misread as “we showed the pathway.” It sits under Chevance as a brake, not as a second keep-mechanism. <https://doi.org/10.1186/s13643-024-02742-7>

Paper 3. Ikäheimo 2018 — keep

The question. Why is the heart-failure residual with official cold days more coherent than the CHD residual with official hot nights? The discussion is haemodynamic, not nocturnal. Afterload is the pressure the heart pumps against. Cold can raise that pressure. A failing heart has little reserve.

What they found. Ikäheimo reviews cardiovascular disease, cold, and exercise. Cold can raise sympathetic outflow, constrict vessels, and increase cardiac workload. The review's own language: a heart-failure patient has little physiological reserve; a failing heart may not be able to compensate. It is a physiology review. It is not this extract.

What our file has. HF is 1.073 per five extra official cold days. All four standard-error constructions exclude 1. Pre-2020 is stronger. Official cold days fall almost entirely in December–February (141 of 145), so the contrast is between winters. We do not measure afterload, blood pressure, or infection.

What we do. Keep the afterload sentence as a hypothesis. Do not name a ventricular phenotype. Do not treat this review as an estimate. <https://doi.org/10.1080/23328940.2017.1414014>

Paper 4. Li et al. 2026 — keep

The question. Same as Ikäheimo: is there a published reason that cold load is a problem for an already compromised heart? This is the second afterload citation, not a new mechanism.

What they found. Li, Wu, Xu and colleagues synthesise cold as a sympathomimetic stimulus. Afterload can rise. Heart failure is named among the clinical risks of decompensation. It is a review. It is not monthly first hospitalisation after a first HF diagnosis in Hong Kong. It does not give a count ratio per five official cold days.

What our file has. Same HF residual as above. A heat-only reading of our paper would miss it. Goggins and Chan's daily Hong Kong HF series is a neighbour question from an earlier decade, not a number to paste into Table 2.

What we do. Keep Li beside Ikäheimo. Two citations, one hypothesis. The live wording already says the present counts do not measure afterload, blood pressure, or infection. Leave that hedge. Do not paste a daily relative risk into Table 2. <https://doi.org/10.3389/fphys.2025.1740919>

Paper 5. Ioannou et al. 2024 — drop

The question. Should the live Discussion keep a confinement study of seven men as an exhibit next to our CHD residual?

What they found. Seven young healthy men, ten days in a chamber. Nights 4–6 at 26.3 °C. Nights before and after at 22.3 °C. Nocturnal core temperature was 0.2 °C higher during *and after* those hot nights. Sleep time fell as overnight core temperature rose. That is a laboratory non-recovery result in young men. Consecutive nights are the experiment.

What our file has. 132 territory-months. People with diabetes or hypertension. First CHD hospitalisation, no admission cause. Outdoor official counts. No core temperature. A monthly total cannot tell consecutive nights from nights scattered through the month. I (count/5) is a reporting scale, not a spell.

What we do. Thinking of dropping, not adding. Hogan can look at the DOI. He is not being asked to vote it onto Table 2. Do not cut the live sentence until he has seen the bullets. <https://doi.org/10.1139/apnm-2024-0105>

DROP: seven men in a chamber are not 132 Hong Kong months

Ioannou et al. 2024	This extract
n = 7 young men 10-day confinement Nights 4–6 at 26.3 °C +0.2 °C nocturnal core T during and after	132 territory-months T2D/HTN first CHD/HF No admission cause Outdoor official counts No core temperature

SCHEMATIC. Protocol numbers from Ioannou et al. 2024; not a coefficient in Table 2.

A chamber of seven men is not this monthly hospital file.

Paper 6. O'Connor et al. 2025 — drop

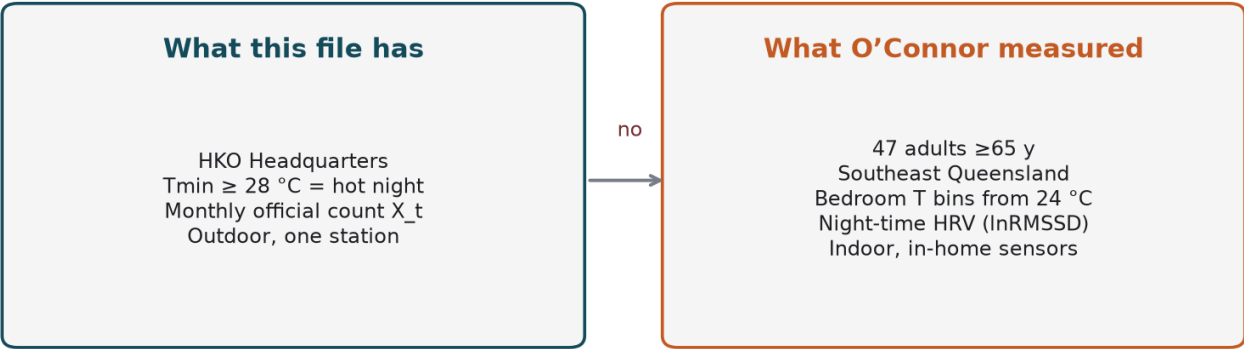
The question. Should we keep a sentence about bedroom temperature above 24 °C and lower heart-rate variability? It is true of that study. It reads as if *this* file measured indoor bedrooms.

What they found. Forty-seven adults aged 65 or older in southeast Queensland, one summer. Bedroom sensors. Night-time heart-rate variability. Relative to nights below 24 °C, warmer bedroom bins had higher odds of a clinically relevant drop in lnRMSSD (about 1.4, then 2.0, then 2.9). Indoor. In-home. Those odds are for a heart-rate-variability drop, not for first hospitalisation.

What our file has. Outdoor official hot nights at one station, $T_{min} \geq 28\text{ °C}$. That is not a 24 °C bedroom. Hogan’s weather paragraph averages Headquarters days to months. It cannot be turned into Queensland bedrooms, or Hong Kong bedrooms. A reader who meets the 24 °C sentence in our Discussion will think we had bedroom sensors. We did not.

What we do. Drop. Indoor night temperature can remain a hypothesis for a later design. It is not a measurement here. Do not add a 24 °C indoor rule to the manuscript. <https://doi.org/10.1186/s12916-025-04513-0>

DROP: outdoor official counts are not bedroom temperature



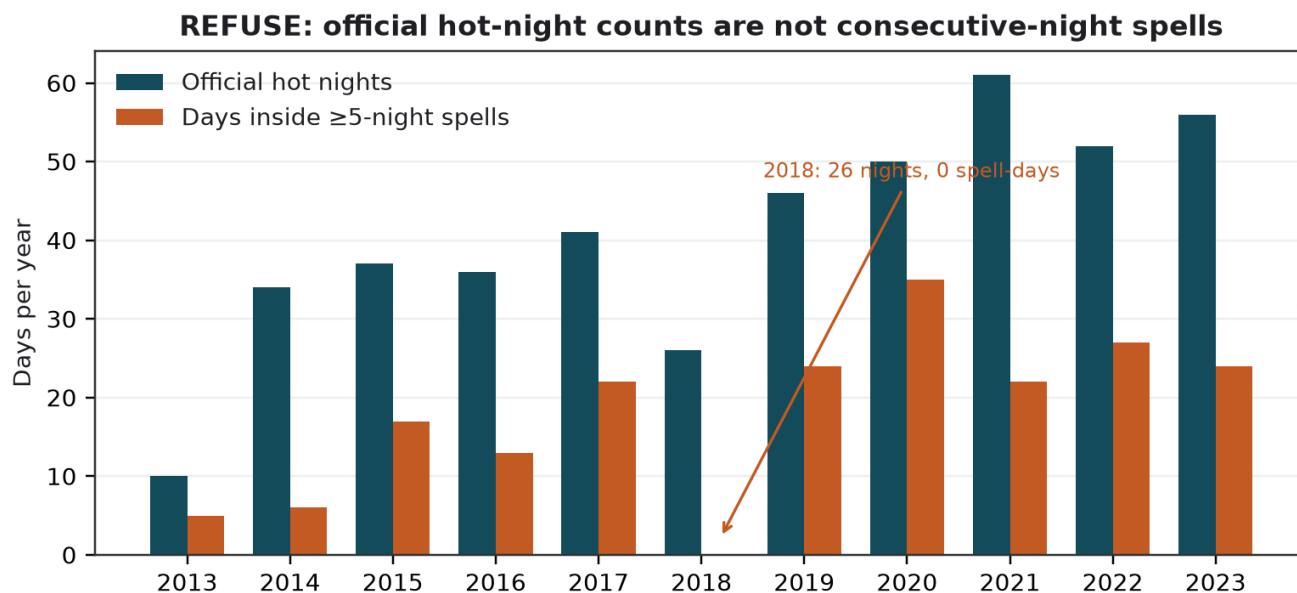
SCHEMATIC. Official hot night is not a 24 °C bedroom.
Outdoor official counts are not bedroom temperature.

Three refusals that are not papers

Indoor temperature from outdoor counts. No. Headquarters Tmin is not a bedroom.

Five extra official days as a consecutive spell. No. $I(\text{count}/5)$ is how we report the monthly count. 2018 had 26 official hot nights and zero days inside a five-night spell. A heavy night-count year need not contain a spell.

Evaluating Hong Kong's heat or cold warnings. No. The estimates do not test the Very Hot Weather Warning or the Cold Weather Warning. They do not count admissions averted.



REAL public HKO Headquarters annual table. A spell-day is a day inside a run of ≥ 5 consecutive official hot nights.

2018 is the exhibit: 26 official hot nights, zero five-night spell-days.